

ML Specification Report — ventilator-pressure-prediction

Ventilator Airway-Pressure Prediction (breath time series) • our from-scratch agent (BiLSTM seq2seq — deep-learning fallback lane)

Figures grounded in the competition's `facts.json` (built by `benchmark_infra/collect_dl_facts.py` from `config.yaml`, `experiments.json`, `cv_result.json`, `STATUS.md`, `headless_run2.log`, `scripts/train.py`), with grading figures from the offline MLE-bench records in `benchmark_results/run2/`. This is a special (deep-learning fallback) lane: grading is offline MLE-bench against a held-out private split — there is no live Kaggle leaderboard, and the framework's DL fields (learning-rate schedule, batch size, epochs, augmentation, transfer learning) are native here rather than translated from GBDT.

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Overview

The task is to predict the airway pressure in a ventilator's respiratory circuit at each of the 80 timesteps of a breath, given the time series of control inputs (lung attributes R and C, commanded inflow `u_in`, expiratory-valve state `u_out`); the metric is **MAE computed only on inspiratory-phase rows (`u_out == 0`), minimized**. The GBDT tree-search pipeline does not apply to per-timestep sequence regression, so the agent fell back to a from-scratch deep-learning recipe: a bidirectional LSTM sequence-to-sequence model trained per breath. The champion is a **BiLSTM-4x256 over 53 per-timestep features, 5-fold ensemble aggregated by median and snapped to the pressure grid**, scoring **CV OOF masked MAE 0.19222** (raw 0.19361) and grading **0.16525** on the offline MLE-bench private split — no medal (bronze needs 0.1364, median is 0.1638), but by far the best of the three benchmarked agents (AIDE 0.34591, NVIDIA 0.40093).

Why it matters. Mechanically ventilating a sedated patient means clinicians hand-tune pressure control against an unseen lung; a model that accurately simulates circuit pressure for a given control

input is a building block for ventilator controllers that adapt to individual lungs, easing clinician load and reducing ventilator-induced injury.

Data

Purpose of Data. Each breath is an 80-timestep time series; the model must output the circuit `pressure` at every timestep, but only the inspiratory phase (`u_out == 0`) is scored — a sequence-to-sequence regression task. **Data Format** is **tabular CSV in long format**: one row per timestep carrying `breath_id`, lung attributes `R` and `C`, `time_step`, `u_in`, `u_out`, and (train only) the target `pressure`; rows group into fixed-length 80-step breaths. **Data Volume**: the graded test split has **603,600 scored rows** (the deliverable submission's verified row count), i.e. 7,545 breaths:

603600 rows / 80 timesteps per breath = 7545 test breaths

The train-split row/breath count was **not recorded** in this lane's structured records (no `eda_summary.json` was produced under the fallback's time budget).

Data Quality. Missing-value / duplicate profiling was likewise **not recorded**. The defining wrinkles are structural rather than statistical: (1) **only the inspiratory phase is scored**, so both the loss and the validation metric must mask expiratory rows; and (2) the target is not truly continuous — training pressures lie on a **950-value grid**, which the pipeline exploits at post-processing. One recorded diagnostic: the five fold models disagree wildly (std 2.57) **only in the unscored expiratory region**; their scored-region spread is 0.115.

Annotation Guidelines. The label is the recorded circuit `pressure` at each timestep; a submission is a real-valued prediction per row, scored by masked MAE — an L1 target that wants the conditional median, which later motivates the median fold-aggregator.

Feature Set. The model consumes **53 per-timestep engineered features** — run1's validated 41 plus 12 new ones ($R \times C$ cross one-hot 9, `u_in` cummax, cummean, reverse-cumsum) — all z-normalized into a `(n_breaths, 80, 53)` tensor:

Group	Features (from <code>scripts/train.py</code>)
Raw inputs	<code>u_in</code> , <code>u_out</code> , <code>time_step</code> , Δt
Lung attributes	$\log_{1p}$ of <code>R</code> , <code>C</code> , $R \cdot C$; one-hots of <code>R</code> and <code>C</code> levels; 9 $R \times C$ cross one-hots
Cumulative / integral	<code>u_in</code> cumsum, time-integral (and <code>C</code> -scaled), cummax, cummean, reverse-cumsum
Local context	lags & leads of <code>u_in</code> at offsets 1–4 with their deltas; finite-difference derivative
Breath-level	<code>u_in</code> mean/max/std/first-value broadcast; <code>u_out</code> cumsum and switch-timing

Splitting strategy. A **5-fold KFold on breaths** (the grouping unit — all 80 timesteps of a breath stay in one fold, preventing within-breath leakage), fold seed 42 (`SEED = 42` in `scripts/train.py`); validation

replicates the scored mask exactly. All 5 folds were trained. Because this is an offline MLE-bench lane there is **no public/private live leaderboard**: the single offline grade (0.16525) is the only external anchor, and it came in better than the honest CV (0.19222) — the direction of that gap is noted honestly but is not diagnosed in the lane's records.

Models & Architecture

Purpose of Architecture. A per-timestep pressure regressor minimizing masked MAE. **Architecture Type** is a **bidirectional LSTM sequence-to-sequence network** — a native deep-learning architecture, not a GBDT translation. The champion is a **5-fold ensemble of a single architecture** (five fold models aggregated by element-wise median), not a heterogeneous blend.

Input Format / Dimension. A `(nBreaths, 80, 53)` float tensor — 80 timesteps $\times$ 53 z-normalized per-timestep features per breath; the sequence axis is real structure (unlike the tabular lanes' 1-D vectors).

Architecture Description. An input projection `Linear`→`LayerNorm`→`SiLU` lifts each timestep into the hidden width, followed by a **4-layer bidirectional LSTM with hidden size 256**, and a 2-layer head emitting one pressure per timestep. The loss is masked L1 (scored region only), so training optimizes exactly what the metric measures. **Model Complexity:** the dense parameter count was **not recorded**; the architecture is fully specified by layers = 4, hidden = 256, bidirectional, over 53 input channels.

Training procedures

Training was a **two-iteration linear ladder** (tree search deliberately skipped — see below), both iterations on the same task and metric:

Stage	Configuration	Masked MAE
run1 (archived)	same BiLSTM recipe, 41 features, budget-bound to 1 fold $\times$ 100 epochs	0.24004 OOF (single fold); offline grade 0.2315
run2 (champion)	+12 features (53 total), bf16 + GPU-resident data, 5 folds $\times$ 260 epochs, median + snap	0.19222 OOF snapped (raw 0.19361)

Fold validation MAEs: 0.19432 / 0.19599 / 0.19258 / 0.18934 / 0.19583.

The **Loss Function** is **masked L1** on `u_out == 0` rows only. The **Optimization Algorithm** is **AdamW** (weight decay `1e-4` in `scripts/train.py`) with gradient-norm clipping at 5. The **Learning Rate** is 0.002 (peak), driven by a native **Learning Rate Scheduler: OneCycleLR over 260 epochs**. The **Batch Size** is **1024 breaths**, with **bf16 autocast** mixed precision.

Training Duration was **~2.6 h** wall-clock for the full 5-fold $\times$ 260-epoch run. The run log credits feasibility to throughput engineering, not architecture: bf16 autocast + the whole dataset resident on GPU + `cudnn.benchmark` cut epoch time from 46.8 s to 6.9 s (~6.8 $\times$), turning run1's "1 fold $\times$ 100

epochs" budget into "5 folds × 260 epochs". The foreground-only constraint was handled by **checkpoint/resume chunking**: 18 sequential ~9-min runs (`--chunk-seconds 520` , `exit 3 = resume`). **Training Memory: not recorded**.

Transfer Learning (native DL field): **none — trained from scratch**; the competition's rules forbid pretrained models (`pretrained_models_allowed: false`), so unlike the other two special lanes there is no pretrained backbone here. **Data Augmentation** (native DL field): **none recorded** in the lane's records.

Reproducibility Standards. Fixed seed 42; per-epoch seeded shuffle replay makes the chunked training **fully deterministic across restarts**; features and per-fold test predictions are cached under `scripts/work/` . **Tree search was deliberately skipped**: one evaluation node here is a full LSTM training run (~30 min/fold, ~2.6 h for an honest 5-fold CV), so a ~60-node search cannot fit the 4 h budget — the skill's documented fallback (single linear iteration, `run1` → `run2`) applied instead.

Inference procedures

Decision Threshold is N/A — a regression metric; raw per-row values are submitted. Two post-processing steps are load-bearing: (1) **median across the 5 fold models' test predictions** — the MAE-optimal aggregator (scored-region fold spread std 0.115; mean-vs-median absolute difference 0.037); then (2) **snap to the 950-value training pressure grid**, validated on OOF (0.19361 → 0.19222, a -0.0014 gain). The submission has 603,600 rows with id order verified against `sample_submission.csv` . **Inference Duration / Inference Memory: not recorded**.

Evaluation & Benchmarking

Task for Performance Evaluation. Predict test-breath pressures on the scored inspiratory mask.

Performance Metrics. Masked MAE (minimize): `run1` 0.24004 (single-fold OOF) → `run2` **0.19222** honest 5-fold OOF (a -0.048 CV improvement); offline MLE-bench grade **0.16525** against thresholds gold 0.1119 / silver 0.1338 / bronze 0.1364 / median 0.1638 — **no medal, worse than median** (0.16525 > 0.1638). The lane's own record notes the medal threshold was "likely still out of reach" and that public solutions place ~0.15 territory at deeper models (5×384), second seeds, and ~300 epochs.

Performance Benchmarking. Grading is **offline MLE-bench** (all three agents scored on the same held-out private split — not a live leaderboard):

Agent	Approach	Masked MAE (offline grade)	Note
Our agent	from-scratch BiLSTM-4x256 seq2seq, 5-fold median + grid snap	0.16525	best of three, no medal
AIDE	(approach not recorded in the grade record)	0.34591	no medal
NVIDIA	(approach not recorded in the grade record)	0.40093	no medal

Our fallback recipe more than halves both comparators' error. The honest caveats: nobody medalled — the recorded medal thresholds sit well below all three scores — and the comparison is grade-only (the comparators' pipelines are not described in the run2 grade records).

Fields marked "not recorded": train-split volume and missing/duplicate profiling (no EDA summary artifact); model parameter count; training peak memory; inference duration and memory; NVIDIA/AIDE approach details. Grading is offline MLE-bench — no public/private live leaderboard exists for this lane.